

CHENGSHANTOU, China

WMO: 547760

Lat: 37.3897N

Lon: 122.6967E

Elev: 47

StdP: 100.76

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.5	-5.0	-13.8	1.1	-2.9	-12.6	1.3	-2.9	19.2	-2.9	17.7	-4.1	10.6	340	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.3	27.6	24.4	26.6	23.8	25.8	23.3	25.6	26.7	24.8	25.9	24.1	25.1	3.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.5	26.4	24.5	19.6	25.6	23.8	18.7	24.8	79.4	26.9	76.2	26.0	73.2	25.1	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.4	14.4	12.6	-8.0	29.7	2.3	1.2	-9.6	30.6	-11.0	31.2	-12.2	31.9	-13.9	32.8		
			-9.1	26.3	1.9	0.9	-10.5	26.9	-11.6	27.5	-12.6	28.0	-14.0	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-0.1	0.8	4.2	8.9	14.3	18.5	21.9	24.0	21.8	16.6	9.5	2.9
	DBStd	8.77	2.96	2.73	2.55	2.51	2.85	2.21	1.87	1.42	1.78	2.81	3.78	3.40
	HDD10.0	1076	312	257	180	49	1	0	0	0	1	55	221	
	HDD18.3	2745	570	491	438	282	128	23	0	1	66	267	479	
	CDD10.0	1813	0	0	1	17	136	256	370	435	354	206	39	1
	CDD18.3	439	0	0	0	0	4	29	112	177	106	12	0	0
	CDH23.3	1249	0	0	0	0	6	38	277	767	161	1	0	0
	CDH26.7	83	0	0	0	0	0	3	25	51	3	0	0	0
Wind	WSAvg	5.9	6.9	6.7	6.5	6.3	5.7	4.9	4.7	4.5	5.1	6.0	6.8	7.0
Precipitation	PrecAvg	751	13	10	19	46	50	76	198	162	91	39	30	17
	PrecMax	1397	51	37	73	136	182	262	932	384	261	106	70	40
	PrecMin	343	1	0	1	2	3	7	15	56	3	9	3	3
	PrecStd	234	11	9	16	37	43	66	168	89	75	27	21	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	8.1	13.1	18.1	24.0	26.2	29.0	29.3	26.9	22.9	18.2	12.0
		MCWB	4.7	4.2	7.6	9.8	14.8	18.6	25.1	26.2	22.4	18.7	14.5	10.2
	2%	DB	6.1	6.2	10.5	15.6	21.8	24.3	27.2	27.9	25.8	21.7	16.9	10.4
		MCWB	4.1	3.8	6.3	9.5	14.2	18.7	24.1	25.0	22.2	18.1	14.0	8.2
	5%	DB	5.1	5.2	8.7	13.8	20.1	22.9	25.9	27.1	24.9	20.9	15.8	8.9
		MCWB	3.3	3.4	5.2	8.8	13.6	18.8	23.5	24.4	21.5	17.5	13.1	6.2
	10%	DB	4.0	4.3	7.4	12.5	18.7	21.7	24.8	26.4	24.2	20.1	14.7	7.6
		MCWB	2.1	2.7	4.7	8.5	13.4	18.5	22.8	24.1	21.1	16.8	11.8	5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	5.9	9.0	12.3	17.6	21.6	26.2	26.9	24.5	20.3	16.4	10.9
		MCDB	6.6	6.7	11.2	15.1	21.5	23.1	27.6	28.3	25.5	21.8	17.4	11.7
	2%	WB	4.6	4.7	7.1	11.2	16.2	20.8	25.1	26.0	23.5	19.3	14.9	8.6
		MCDB	5.7	5.5	9.0	13.2	18.7	22.3	26.3	27.1	24.6	21.0	16.3	10.1
	5%	WB	3.5	3.9	6.2	10.3	15.4	20.1	24.1	25.4	22.7	18.5	13.5	6.8
		MCDB	4.8	4.7	7.6	12.1	17.8	21.4	25.2	26.4	24.0	20.4	15.3	8.3
	10%	WB	2.4	3.0	5.4	9.5	14.6	19.4	23.3	24.7	22.0	17.5	12.3	5.4
		MCDB	3.8	4.1	6.8	11.3	17.0	20.8	24.2	25.7	23.5	19.6	14.6	7.3
Mean Daily Temperature Range	5% DB	MDBR	3.5	3.6	4.1	4.2	4.6	3.9	3.4	3.3	3.3	3.5	3.8	3.7
		MCDBR	4.4	4.7	6.2	6.6	7.1	5.7	5.0	4.3	4.1	3.9	4.3	4.7
		MCWBR	4.3	3.9	4.1	3.9	3.8	3.3	3.1	2.7	2.9	3.6	4.3	4.7
	5% WB	MCDBR	4.1	4.3	5.3	5.3	5.8	4.5	4.2	3.7	3.5	3.5	4.2	4.5
		MCWBR	4.6	4.0	4.2	3.9	3.8	3.1	2.9	2.5	2.8	3.8	4.8	5.0
Clear-Sky Solar Irradiance	taub	0.462	0.514	0.570	0.603	0.629	0.707	0.693	0.603	0.531	0.553	0.531	0.471	
	taud	1.885	1.801	1.706	1.693	1.688	1.597	1.683	1.865	1.964	1.805	1.803	1.880	
	Ebn at Noon	667	687	694	704	696	642	647	697	723	642	590	620	
	Edh at Noon	149	183	221	236	241	264	241	196	168	180	161	141	
All-Sky Solar Radiation	RadAvg	2.20	3.05	4.33	5.11	5.57	5.25	4.65	4.75	4.37	3.40	2.35	1.88	
	RadStd	0.18	0.23	0.34	0.33	0.46	0.48	0.51	0.52	0.34	0.25	0.23	0.12	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page